

GEN AI IN BUSINESS

Kerry Back

Rice Business

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Today's deck

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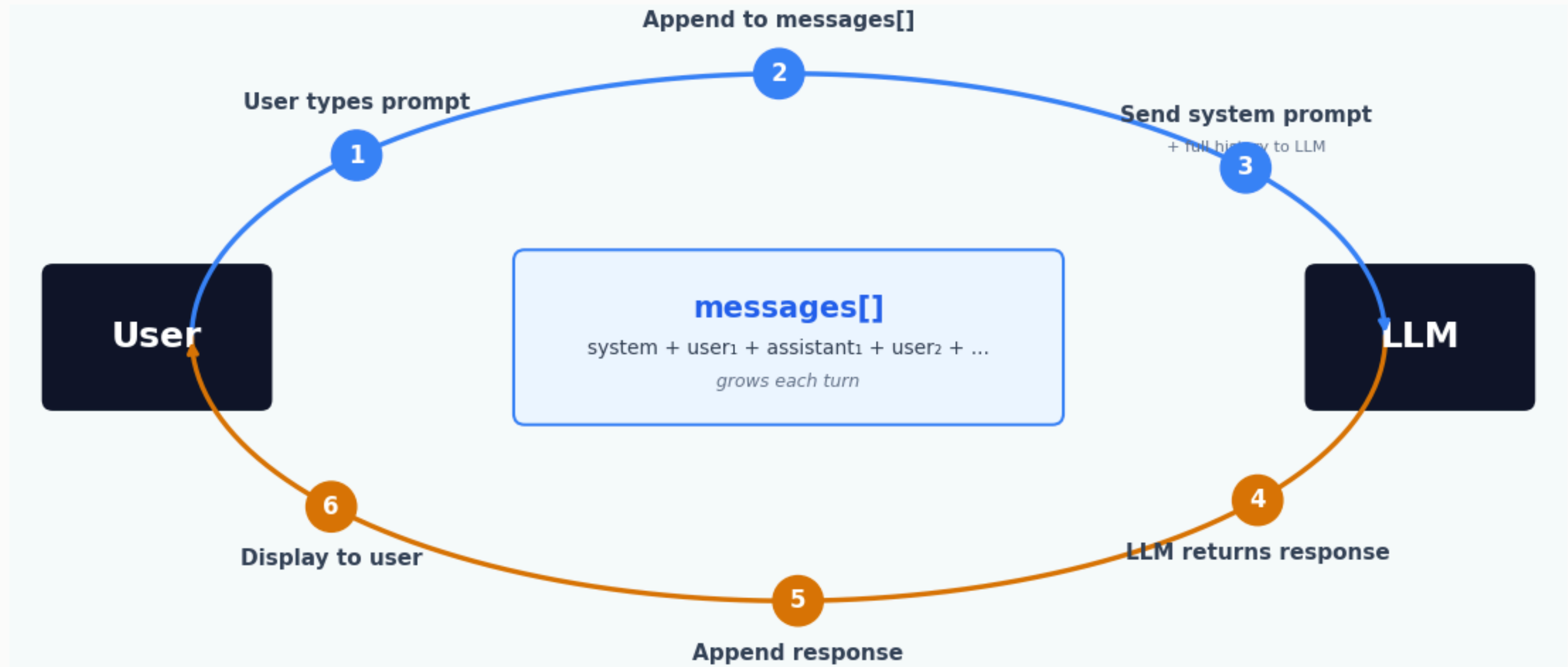
How AI is used in business

USE	WHAT IT LOOKS LIKE
Chatting about corporate documents	Retrieval-augmented generation
Personal assistants	Calendar, email, notes, scheduling
Data access and analysis	Query the database, build the chart
Document preparation	Memos, decks, reports, filings
Customer support	Phone, email, chat, triage of tickets
Financial trading	Signals from news, filings, transcripts
Software development	Writing, reviewing, and testing code
Marketing and content	Copy, campaigns, personalization
Contract and legal review	First-pass review, clause extraction

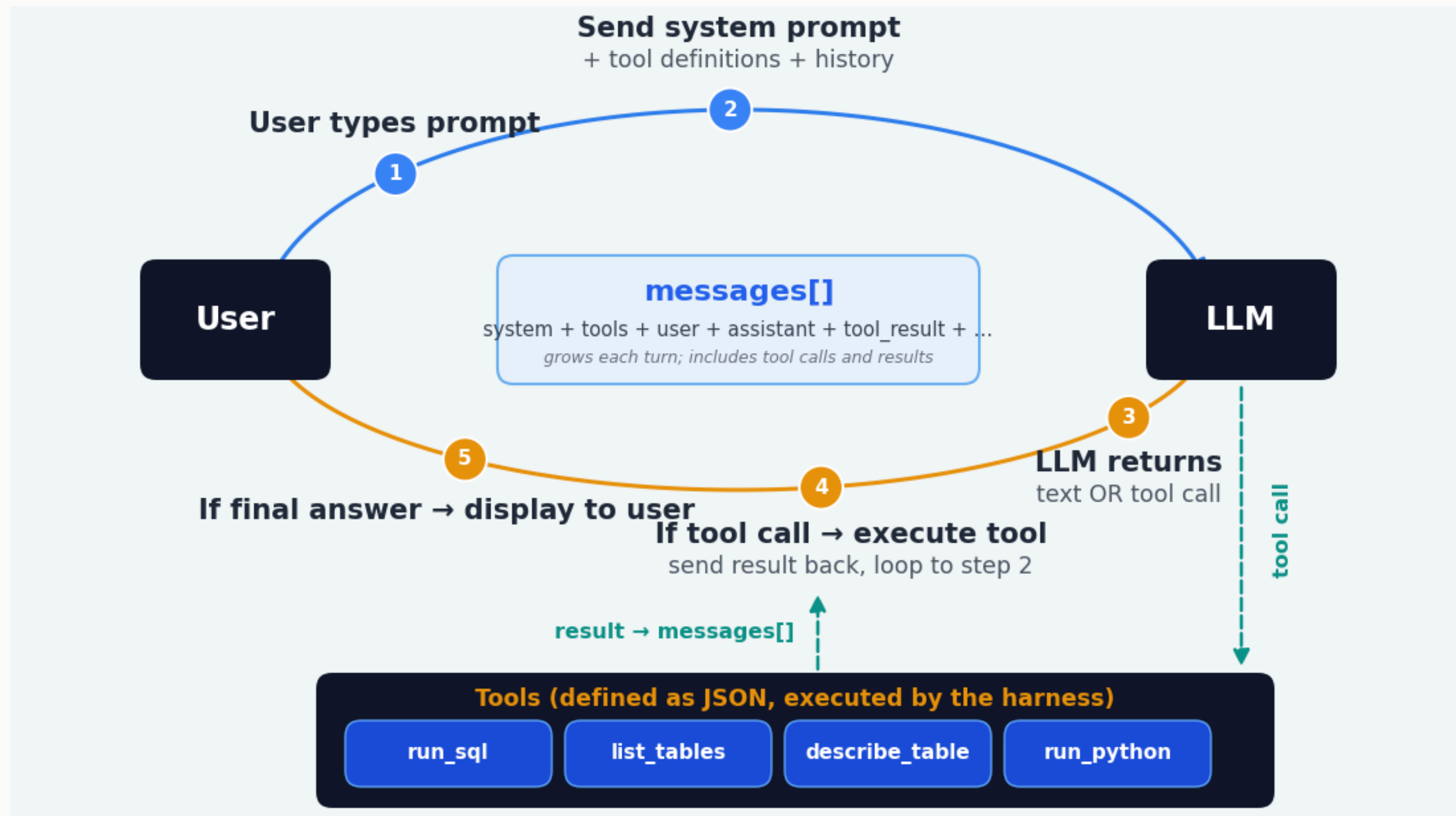
HOW CHATBOTS AND AGENTS WORK

A chatbot returns text. An agent is a chatbot with tools it can ask to use.

ANATOMY OF A CHATBOT



ANATOMY OF AN AGENT



An agent has three parts

The model

Decides. Emits text: sometimes an answer, sometimes a request to use a tool.

The harness

Ordinary software wrapped around the model. Runs the tool the model asked for.

The tools

The things that touch files, databases, and other systems.

The model never runs anything itself. It chooses tools to use. The harness executes, and the harness returns information about the tool outcome to the model.

Read and write files

Documents, spreadsheets, code.

Run code

Python, SQL, the command line.

Search and fetch the web

Run a search, then download and read a page.

Use a screen

Click and type in a browser or a desktop app.

Call a service

A database or another program, even another agent.

Start another agent

A subagent with its own instructions and tools.

What the tools are

THE CLAUDE CODE AGENT

The tools of Claude Code

TOOL	WHAT IT DOES
Read, Write, Edit	Open and change files — documents, spreadsheets, code
Bash	Run commands and programs on the machine
Glob, Grep	Find files, and search inside them
WebSearch, WebFetch	Run a search, then download and read a page
Task	Start a subagent with its own instructions
MCP connectors	Call an outside service: a database, an API

Written instructions for a specific task, together with the code scripts and other files that task needs.

EXAMPLE	WHAT IT IS FOR
<code>xlsx</code>	Excel workbooks
<code>pptx</code>	PowerPoint decks
<code>docx</code>	Word documents
<code>frontend-design</code>	Web pages and interfaces
<code>finance-data</code>	Market and economic data
The first four come from Anthropic. <code>finance-data</code> is one Claude wrote for me.	

Skills

Sign in to AI for demos

- 1 Go in browser to `lab-oct.rice-business.org`
- 2 Username = `firstname_lastname` — hyphenate a surname that has more than one part
- 3 Password = `jgsbai`

Environment for this session

What runs

Claude Code running
Claude Sonnet

What is installed

Supporting software
(Python, JavaScript),
Python libraries, Claude
plugins

What you see

File browser (download and
upload), terminal, local
browser view (View tab)

Everything running in the cloud, billed to the school.

WHERE	WHAT
Right panel	Claude Code. The prompt window is at the bottom.
Left panel	File Browser (default), Terminal, View. Ignore Terminal and View for now.
Three-dots menu on left	Upload (from your laptop) and Download (to your laptop).

Workspace

1. Build an Excel workbook

Prompt

Build an Excel workbook containing a mortgage amortization table. Include a chart of the amount going to interest and the amount going to principal each month.

Use the **Refresh** button above/right of the left pane to see the new file in the File Browser. MS Office is not installed here — download Office documents from the three dots menu and open them on your laptop.

2. Data merging, filtering, aggregation, ...

Five spreadsheets from a fictitious company — customers, orders, order details, products, categories — are in the data folder.

PROMPTS

- 1 Which countries generated the most revenue? Compute revenue as price times quantity.
- 2 Create a bar chart of revenue by country for the top five countries.

The model writes code to merge orders with order details and then merge with customers, then add revenue across all order details of orders from customers in each country.

3. Software development

Tell Claude you want a single-page HTML app that allows the user to input mortgage terms and computes and displays the amortization table and a chart of the amount going to principal and interest each month.

Refresh and locate the html file it creates in the lab's file explorer. Then download, locate in your File Explorer (Mac Finder), and double click.

Tell Claude you want to compute daily stock returns for Nvidia since Jan 1 using Yahoo Finance data and produce a histogram.

4. Get data

HOW LARGE LANGUAGE MODELS WORK

Enough of the mechanism to explain what these models are good at and where they fail.

Tokenization

Text is cut into pieces from a fixed vocabulary before the model sees any of it.

"The cat sat on the mat" → ["The", " cat", " sat", " on", " the", " mat"]

"Unbelievable" → ["Un", "believ", "able"]

Common words stay whole; rare and long words break into pieces. A typical vocabulary runs to about 100,000 tokens.

Numbers break on the same rules, which have nothing to do with arithmetic. 13,963,378.65 reaches the model as a handful of unrelated pieces.

Vectors

Each token is associated with a vector, meaning a list of numbers. So, a sequence of text becomes a sequence of lists of numbers.

Each token is a list

“king” becomes a list of 4,096 numbers.

“queen” becomes a different but somewhat similar list of 4,096 numbers.

Tokens used in similar contexts end up near each other.

The space is mostly empty

There are far more possible vectors than there are tokens, so most points in the space name nothing.

The prediction can land between tokens. The “closest” token is usually selected but not always (so LLM output is somewhat random).

The whole task

Given the tokens (lists of numbers) so far, predict the token (list of numbers) that comes next.
That is the entire operation.

Everything a model appears to do — answering, summarizing, writing code, refusing — is that one prediction, repeated.

A neural network is a function

THE IDEA

$y = 4x$ is a function. The 4 is a parameter.

Change it to 3 and you have the same structure with different behavior.

THE SCALE

Same idea with perhaps trillions of parameters, arranged as a transformer — the architecture published by Google researchers in 2017.

The pipeline

- 1 Tokenize the input.
- 2 Look up the vector for each token.
- 3 Feed the vectors into the network.
- 4 The network produces one output vector.
- 5 Find which token vectors are closest to it.
- 6 Emit a token, append it to the input, and start again at step 1.

A model trained only to predict the next token continues your text. It does not answer a question.

- 1 Role tokens. The text is wrapped in markers: `<user>` your question `<assistant>`.
- 2 Instruction tuning. The model is further trained on examples where what follows `<assistant>` is a useful reply.
- 3 Reinforcement learning from human feedback. Raters score replies, and the model is trained toward what raters preferred.

The mechanism did not change. What changed is which continuation is the likely one. GPT became Chat-GPT in 2022.

From prediction to conversation